HDOC DAY 52

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named previous_greater() using its address. Use pointers to pass the array between the calling function and previous_greater(). The function previous_greater() must:

- Find the previous greater element for each element in arr[].
- Search only the elements to the left of the current element and select the nearest element that is greater than the current element.
- Display -1 if no greater element exists to the left.
- Display the results in the order of their appearance in the array, separated by commas.
- Use a nested-loop approach to solve the problem.

CODE

```
#include <stdio.h>


void previous_greater(int *arr, int n)
{
   int i, j, found;


   for (i = 0; i < n; i++)
   {
      found = 0;


      for (j = i - 1; j >= 0; j--)
      {
         if (*(arr + j) > *(arr + i))
```

```
        {
            printf("%d", *(arr + j));
            found = 1;
            break;
        }
    }

    if (found == 0)
    {
        printf("-1");
    }

    if (i < n - 1)
    {
        printf(",");
    }
  }
}

int main()
{
  int n, i;

  printf("Enter the size of the array: ");
  scanf("%d", &n);

  int arr[n];
```

```
    printf("Enter the array elements: ");
    for (i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }

    previous_greater(arr, n);

    return 0;
}
```